



PORTFOLIO EDITION

BLUE TEAM INTERNSHIP

Week 11 Task Report

Email Forensics · Host Artefact Deep-Dive

Super Timeline Analysis

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01. Executive Summary

This report documents Week 11 of the Blue Team Internship, covering Days 74 through 80. The week's brief centres on Jean, CFO of M57.biz, and requires correlating email evidence, host-based artefacts (LNK, Prefetch, registry), and a filtered super timeline to determine whether a confidential spreadsheet (m57plan.xls) was intentionally leaked or sent to its legitimate recipient in error.

The evidence image for this investigation, provided as a split EnCase image (nps-2008-jean.E01 and nps-2008-jean.E02), could not be downloaded successfully this week. The shared link returned an 'Unauthorized' access error rather than a corrupted or incomplete file, meaning this is a permissions issue on the sharing end rather than a local download or tool problem. Access has been requested and the reviewer has been notified directly; this is a separate case file from the Mantooth.E01 image used in Week 10, whose earlier access issue was independently traced to a corrupted download and has since been resolved with a verified working copy.

In place of case-specific findings, this report answers every task across Part A (email extraction and triage), Part B (LNK, Prefetch, and registry deep-dive), and Part C (super timeline generation and filtering) at the methodology level: what each task requires, exactly where in the evidence the answer lives, and the precise technique to extract it. No email addresses, timestamps, or file paths specific to this case are asserted, since none were extracted from a working case this week.

Task Block	Status	Notes
Days 74-75 - Part A: Email Extraction & Triage	Blocked - methodology documented	Image access pending; see Section 06
Days 76-78 - Part B: Host Artefact Deep-Dive	Blocked - methodology documented	Dependent on Part A image access
Days 79-80 - Part C: The Super Timeline	Blocked - methodology documented	Dependent on Parts A and B
Section 06 - Technical Access Issue Log	Complete	Permission error documented and escalated to reviewer

02. Week 11 Objectives

By the end of Week 11, the objective is to be able to:

- Locate and parse email archive formats (PST, OST, MBOX, EML) from a forensic image.
- Read raw email headers and distinguish a To field's display name from the actual underlying recipient address.
- Corroborate email evidence with host-based artefacts: LNK creation timestamps, Prefetch execution records, and NTUSER.DAT registry writes.
- Generate a super timeline of system events and apply the Start Big, Filter Down methodology to isolate a precise time window.
- Synthesise evidence across email, host artefacts, and timeline into a single, defensible investigation conclusion.

CASE PARAMETERS (from task brief)

Target file: m57plan.xls, Jean's desktop. MD5: e23a4eb7f2562f53e88c9dca8b26a153. Last modified: 2008-JUL-20 01:28:03 GMT. Investigation window: 01:25:00-01:30:00 GMT. Case timezone: GMT throughout.

03. Part A - Email Extraction & Triage (Days 74-75)

Objective: locate Jean's email archives, isolate the outbound email sending m57plan.xls, and determine - from the raw header, not the client's display - the actual delivery address the file was sent to.

#	Task / Question	Hive / Tool / Location	Methodology
1	Add split image (.E01 + .E02) as a single data source; set case timezone to GMT.	Autopsy -> Add Data Source -> Disk Image or VM File.	Keep both .E01 and .E02 in the same folder and point the wizard only at the .E01 file - the tool auto-detects and stitches the .E02 segment. Set the case time zone to GMT/UTC before ingest so all timestamps align with the parameters given in the brief.
2	Locate all email archive files and document path/size for each.	Keyword/file search across the image for extensions .pst, .ost, .mbox, .eml.	Autopsy's file search can be filtered by extension. The archive's path indicates the client: a local .pst under a user profile suggests Outlook with local storage; an .ost suggests a cached Exchange mailbox; .mbox suggests Thunderbird; standalone .eml files indicate individually saved messages.
3	Parse the archive(s) and isolate emails sent 20 July 2008.	Autopsy's built-in email module (Data Artifacts -> E-Mail Messages), or export and open in a dedicated tool if native parsing is incomplete.	Sort/filter the parsed message list by Date Sent to isolate the target day before doing anything else, per the brief's own guidance not to skip this step.
4	Locate the outbound email attaching m57plan.xls near 01:28:03 GMT; document date/time, To display name, sender display name, subject, attachment name/size.	Within the 20-July message list, filter for outbound messages with an attachment.	Identify the single message whose sent timestamp is closest to the file's last-modified time (01:28:03 GMT) and whose attachment matches m57plan.xls by name/size. Record every requested field directly from the parsed message view first.
5	Determine the actual recipient address (not the display name) and state whether it matches Jean's account of events.	View raw message source / headers - in Autopsy, the 'view raw message' or source option on the email artifact; look for the To:, Delivered-To:, or Received: header lines.	The parsed 'To' field in any client only shows a display name, which the sender's own address book can set to anything. The raw header's actual email address is what the mail server used to route the message. Document both values side by side and state explicitly whether they match - this comparison is the pivot finding of Part A.
6	Review all other emails on 20 July 2008 for relevance to the President or any third party.	Same E-Mail Messages artifact list, full day.	Read every message sent/received that day, noting any correspondence involving the company President, any external/competitor domain, or any reference to the target file or confidential data.

Understanding Verification - Part A

A To field's display name is a label the sender's mail client stores locally and can be edited freely - it does not participate in message delivery. The actual recipient address is a separate structured field the mail transfer protocol uses to route the message to a mailbox; it cannot be displayed incorrectly the way a name can. When they diverge, the display name reflects what the sender believed or intended, while the address reflects what actually happened to the message - and in an insider-threat investigation, the address is the fact that matters, because it proves where the data actually went regardless of who the sender thought they were addressing.

04. Part B - Host-Based Artefact Deep-Dive (Days 76-78)

Objective: corroborate the email evidence with independent, harder-to-fabricate host artefacts, all evaluated against the anchor timestamp of 01:28:03 GMT.

LNK File Analysis

#	Task / Question	Hive / Tool / Location	Methodology
1	Locate the LNK shortcut for m57plan.xls in Jean's Recent folder.	Documents and Settings\[username]\Recent\ (Windows XP-era path).	Browse directly to the XP-era Recent path in the file tree, since this image predates the later Windows path structure.
2	Document creation timestamp, last-accessed timestamp, target path, volume info, machine name.	Parse the LNK with Autopsy's Recent Documents artifact or an external LNK parser (e.g., LECmd).	A LNK file is written the moment its target is first opened, so its creation timestamp is direct evidence of when Jean opened m57plan.xls. The embedded volume serial number and machine name can additionally confirm the shortcut originated from this specific machine.
3	Compare LNK creation time to the 01:28:03 GMT anchor.	Direct comparison against the case parameter.	Document the delta in seconds/minutes. Close alignment supports Jean's account that she opened the file around that time; a large or unexplained gap would need separate investigation (see the tunnelling note below).

Prefetch Analysis

#	Task / Question	Hive / Tool / Location	Methodology
1	Locate Excel's prefetch file.	Windows/Prefetch/EXCEL.EXE-<8-char hash>.pf.	Search the Prefetch directory for the EXCEL.EXE-prefixed file; on XP-era systems the naming convention is executable name + hyphen + 8-character hash.
2	Document last execution time, run count, files/directories accessed.	Parse with Autopsy's Recent Activity module or an external tool (e.g., PECmd).	The prefetch file's header stores the last-run timestamp and run count; its file-reference section lists paths Excel touched on that run, which should include m57plan.xls itself.
3	Determine whether prefetch execution precedes, matches, or follows LNK creation.	Direct comparison of the two extracted timestamps.	Excel must launch before it can create/update a LNK entry for a file opened within it, so the expected order is prefetch execution at or slightly before LNK creation. Document the actual observed order and what it implies about the user's action sequence.

Registry State

#	Task / Question	Hive / Tool / Location	Methodology
1	Locate the RecentDocs entry for m57plan.xls.	NTUSER.DAT - Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs (XP-era key path).	Export NTUSER.DAT from the user's profile in Autopsy and parse with Registry Explorer or RegRipper; locate the m57plan.xls entry and its key last-write timestamp.
2	Identify any other NTUSER.DAT keys last written within 01:25:00-01:30:00 GMT.	Full NTUSER.DAT key enumeration, sorted/filtered by LastWrite time.	Registry Explorer can sort all keys by last-write time; filter to the five-minute investigation window to catch any other user-driven registry activity clustered around the event, not just the RecentDocs key.

#	Task / Question	Hive / Tool / Location	Methodology
3	Cross-reference LNK, Prefetch, and RecentDocs timestamps in one table.	Synthesis of the three artefacts above.	Build a three-row table (artefact / timestamp / user action represented) and assess whether all three cluster tightly around 01:28:03 GMT, which would corroborate Jean's account, or reveal a gap that needs separate explanation.

Understanding Verification - Part B

A single artefact - even a precise one - can in principle be altered or coincidental. What makes LNK creation, Prefetch execution, and a RecentDocs write together strong evidence is that they are recorded by three independent OS subsystems, at essentially the same moment, without any of them being aware of the others. Fabricating a consistent story across all three would require modifying multiple unrelated binary structures in a mutually consistent way - which is far harder than editing a single log - so agreement across independent artefact categories is treated as much stronger evidence than any one category alone.

Note on Windows file system tunnelling: if the LNK timestamp appears earlier than expected, this can result from tunnelling, a Windows mechanism that preserves original timestamps of recently deleted-and-recreated files with the same name. Any such discrepancy should be documented and explained as a known behaviour rather than silently omitted.

05. Part C - The Super Timeline (Days 79-80)

Objective: generate a full system timeline, apply the Start Big, Filter Down methodology, and reconstruct the second-by-second sequence around the anchor event.

#	Task / Question	Hive / Tool / Location	Methodology
1	Generate a comprehensive system timeline (file system MAC times, registry writes, event logs).	Autopsy's Timeline feature, or log2timeline/plaso if a dedicated super-timeline tool is available.	Export the full timeline as CSV to a working folder. Expect hundreds of thousands of entries - this is expected and not itself a problem to solve.
2	Apply a date filter to 20 July 2008 only.	Timeline tool's date-range filter.	Narrow to the single day before applying any other filter, to avoid false matches from unrelated dates elsewhere in the timeline.
3	Apply the time window filter, 01:25:00-01:30:00 GMT.	Timeline tool's time-range filter, confirmed to be displaying GMT.	Verify the tool's display timezone is GMT before filtering - a mismatched timezone is the most common reason this filter returns nothing.
4	Apply a keyword filter for 'm57plan' within the filtered window.	Timeline tool's keyword/text filter, searching the full path field.	Ensure the filter searches full paths, not just an isolated filename field, since the string may only appear mid-path depending on how the tool indexes entries. This should leave a small, reviewable set of entries.
5	Reconstruct the second-by-second sequence; identify the four core events (registry update, prefetch write, LNK creation, email send/compose record).	Manual review of the final filtered result set.	For each remaining entry, document timestamp (to the second, GMT), event type, artefact/path, and an interpretation of the user action or system process responsible. Confirm the four core events appear in a chronologically coherent order.

Sequence of Events - Table Template

To be completed once the timeline is generated against a working image. Template structure below.

Timestamp (GMT)	Event Type	Artefact / Path	Interpretation
[HH:MM:SS GMT]	Registry update	NTUSER.DAT / RecentDocs	Excel opened / document referenced
[HH:MM:SS GMT]	Prefetch write	EXCEL.EXE-<hash>.pf	Excel execution recorded
[HH:MM:SS GMT]	File system / LNK creation	Recent\m57plan.xls.lnk	Shortcut created on first open
[HH:MM:SS GMT]	Email record	Email archive send/compose entry	Message dispatched with attachment

Understanding Verification - Part C

A full timeline of a Windows system routinely contains hundreds of thousands to millions of timestamped events, and reading it sequentially does not scale to any real investigation. The Start Big, Filter Down method works because each successive filter (date, then time window, then keyword) is anchored to something already known with confidence - the day, the narrow window around a specific file's last-modified time, and the file's own name - so each filter step discards volume without discarding relevance. The alternative, browsing an unfiltered timeline hoping to notice the right entries, does not have this guarantee and does not scale as image size grows.

06. Anticipated Investigative Significance

Although no case-specific findings could be extracted this week, it is worth setting out clearly what the completed investigation is expected to show and why it matters, so the eventual findings can be evaluated against a stated hypothesis rather than assembled without direction once access is restored.

What a header-address mismatch would prove

If the raw header's actual recipient address for the m57plan.xls email differs from the display name Jean believed she was sending to, that is close to direct evidence of a misdirected send rather than a deliberate leak - a display name can be wrong for entirely innocent reasons (an outdated address book entry, a similarly-named contact), and the raw address is what the mail server actually used. Conversely, if the two match, the investigation would need to look elsewhere - Jean's own account of events - for how the file reached an unintended party.

What the host-artefact corroboration is actually for

The point of cross-checking LNK, Prefetch, and registry timestamps against 01:28:03 GMT is not to confirm the file was opened - that much is already implied by its last-modified timestamp. It is to establish whether Jean's own subsequent actions on the machine (opening the file, running Excel) are consistent with an ordinary, momentary lapse around the time of sending, or whether a longer gap or an inconsistent sequence suggests a different, more deliberate sequence of events that would need a separate explanation.

What the super timeline is expected to add beyond the other two parts

Email headers and host artefacts each tell part of the story in isolation; the super timeline's value is in showing whether every other system event in the same five-minute window is consistent with an ordinary send-and-forget action, or whether something else - another file access, a second email, an unusual process - occurred in the same window that neither the email evidence nor the host artefacts alone would surface.

07. Technical Access Issue & Troubleshooting Log

Consistent with this internship's reporting standard, this section documents the access issue encountered this week rather than omitting it.

Issue

The shared-drive link provided for nps-2008-jean.E01 / nps-2008-jean.E02 returned an 'Unauthorized' / access-denied response rather than allowing download. This is distinct from the Week 10 issue (Mantooth.E01), which was subsequently traced to a corrupted/incomplete local download rather than a permissions problem - a working copy of that Week 10 image has since been obtained and verified separately.

Steps Attempted

- Opened the shared link directly in-browser and confirmed the 'Unauthorized' message rather than a generic load failure.
- Used the 'Request Access' option where available and submitted an access request under the correct account.
- Escalated directly to the assigned reviewer, describing the exact error and requesting either approval of the pending access request or an alternative working link.
- Confirmed this is a distinct evidence file from Week 10's Mantooth.E01, so the earlier resolution (obtaining a clean copy) does not apply here without a new, authorised link for this specific case.

Current Status

Access request pending reviewer approval at time of submission. This report proceeds on a methodology basis for all three parts so the full investigative approach is documented and ready to execute against nps-2008-jean.E01/.E02 as soon as access is granted.

08. Key Learnings & Reflection

The most consequential finding in an email case is often one field, read correctly

This week's brief is built around a single distinction - display name versus actual delivery address - that is trivial to state but easy to skip if an analyst trusts what the email client shows on screen. The broader lesson is that forensic tools frequently present a convenient summary view alongside a raw, authoritative one, and knowing which one a specific question actually requires is itself a skill, not just a matter of clicking through a parser.

Corroboration is a methodology, not a formality

Cross-checking LNK, Prefetch, and registry timestamps against each other is not busywork added after the 'real' finding - it is what turns a single artefact into a defensible claim. Independent artefact categories that agree close off the argument that any one of them was coincidental or fabricated.

A permissions error and a corrupted file require different fixes

Week 10's blocker and this week's blocker looked similar from the outside ('the file won't open') but had different root causes - one was a bad download, the other is an access grant that only the file owner can resolve. Diagnosing which situation is actually occurring, rather than repeating the same fix, is what got Week 10 unblocked and is the same discipline being applied to Week 11's separate issue here.

09. Closing Statement & Sign-Off

This concludes the Week 11 deliverable report for the Blue Team Internship, covering the planned email forensics, host-artefact corroboration, and super timeline analysis of the Jean/M57.biz evidence image. Image access for nps-2008-jean.E01/.E02 was blocked by a shared-drive permission error, documented in Section 06 and escalated to the reviewer. In its place, this report provides a complete, task-by-task methodology for all three parts of the week, so the investigation can be finished directly against the evidence once access is granted.

SUBMITTED BY

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TRACK

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REPORTING PERIOD

Week 11 - Days 74-80